

Abstract:

KANDO is a completely decentralized social network with free gas (no payment). This platform is based on hexagons similar to the structure of a beehive. From the center of the hive, each cell has six neighbors. There is one user in each cell. The location of these people is determined by the coordinate axis (q, r) . In this system, content contagion is based on the rule of 3 confirmations derived from Damon Centola's theory of complex contagion. The system works in such a way that a message is first sent to six neighboring cells and after receiving at least three confirmations (apart from the sender), the message is distributed to other neighboring cells. If the minimum number of votes is not obtained, the news is no longer distributed. This function is used to prevent the spread of fake news and rumors.

In the KANDO Co-Growth Protocol, we have three phases for entering citizenship: the names of these three phases are: Reserved (temporary), Candidate (being confirmed), and Citizen (confirmed). A person can achieve citizenship when every six neighboring cells have at least six cells. This method, unlike proof of work or staking, provides resistance to cyber attacks. All identity information is stored in a distributed hash table (DHT). Without the need for blockchain and cryptocurrencies and imposing additional costs such as gas fees for ordinary users. The KANDO hexagonal topology has been submitted to the Estonian Patent Office as a utility model, but the entire project will be released as open source under the AGPLv3 license.

1. Introduction: Today's Internet and applications are largely centralized, which can lead to risks such as censorship, collection of personal data, and in the event of an Internet outage, they officially enter a single point of failure. Decentralized alternative apps also often have gas fees, low processing power, and are vulnerable to cyber attacks and spam because they are still based on simple contagion (one Click = one Share = one Contagion). Research by Professor Damon Centola of the University of Pennsylvania has shown that the spread of cooperative trust and social actions ultimately leads to the spread of behavior, following a complex contagion pattern. Adoption of a new behavior requires reinforcement from multiple independent sources.

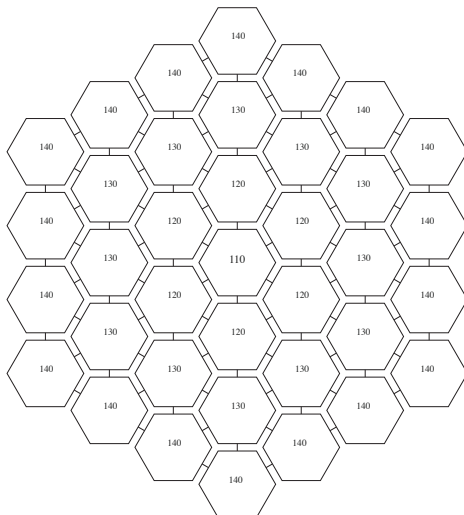
KANDO has transformed this scientific insight into a practical network architecture that is a modern and novel innovation.

2. Problem: Existing decentralized networks suffer from four technical drawbacks: 1- Uncontrolled message flooding: Content reaches all connected nodes without regard to quality, wasting bandwidth and facilitating the spread of misinformation 2- Lack of local trust: Cyberattacks are cheap. Because identity is not tied to any structural requirements, global consensus is also slow and expensive. 3- Lack of spatial embedding: Most P2P graphs are random, making community-based routing difficult without external metadata. 4- High transaction costs: Blockchain-based solutions impose a transaction fee on the user, preventing mass adoption.

KANDO solves these problems intrinsically due to its structure, without the need for cryptocurrencies or transaction fees.

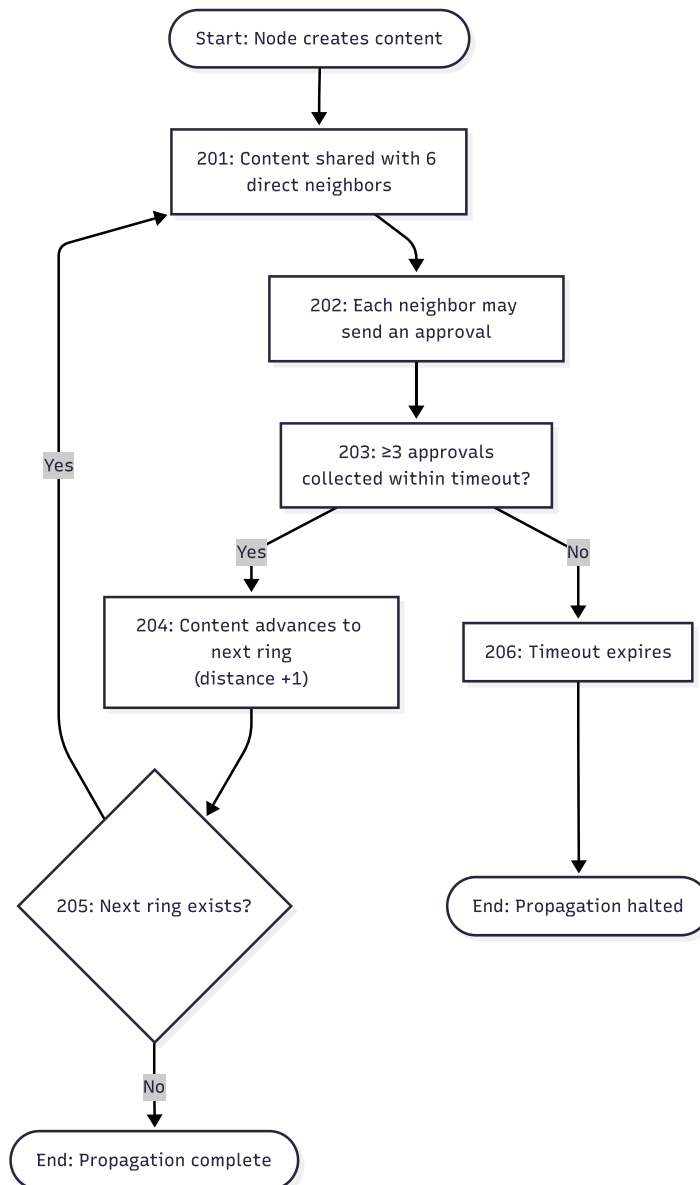
3- Main architecture:

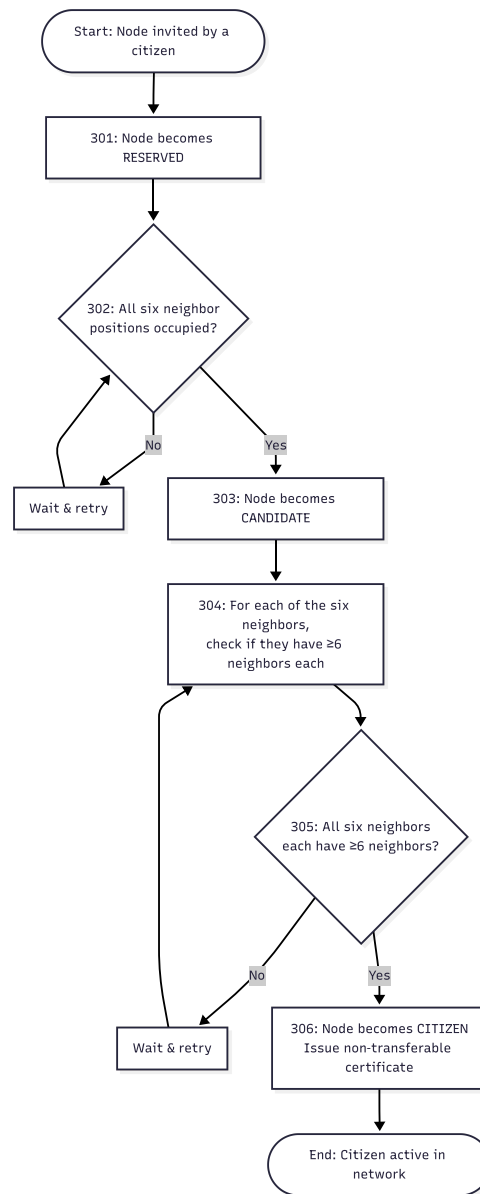
3-1 Hexagonal topology and axial coordinates: Each node is equipped with a pair of axial coordinates (q, r). The implicit third coordinate will be ($s = -q - r$). The hexagonal distance between two nodes is calculated as follows $\Rightarrow$ (Distance = $(|\Delta q| + |\Delta r| + |\Delta s|) / 2$). This metric defines the rings. Ring 0 contains the cell or central node (queen) at (0,0). Ring 1 contains six cells at distance 1, each of which has exactly three neighbors (queen and two adjacent neighbors in ring 1). Ring 2 contains 12 cells (nodes). Ring 3 contains 18 nodes, and so on. Each node maintains a list of up to six direct neighbors. Why the number six? A hexagonal network maximizes the clustering coefficient (between 0.4 and 0.67) compared to square or random graphs, which is optimal for complex contagions.



3-2 Rule of 3 Confirmations: (Complex Contagion) When a citizen creates content, that content is initially shared with only his or her six direct neighbors. Each neighbor can send a cryptographic confirmation. The source node counts the confirmations. If at least three independent confirmations are collected in a given time interval (e.g. 24 hours), the content is propagated to the next ring. This process is repeated in other rings, and if the threshold is not met, the propagation is stopped permanently. This mechanism provides the following advantages: Bandwidth reduction: Poor-quality content is stopped in the first ring, reducing network traffic by 60–80% compared to flooding or gossip protocols. Inherent spam filter:

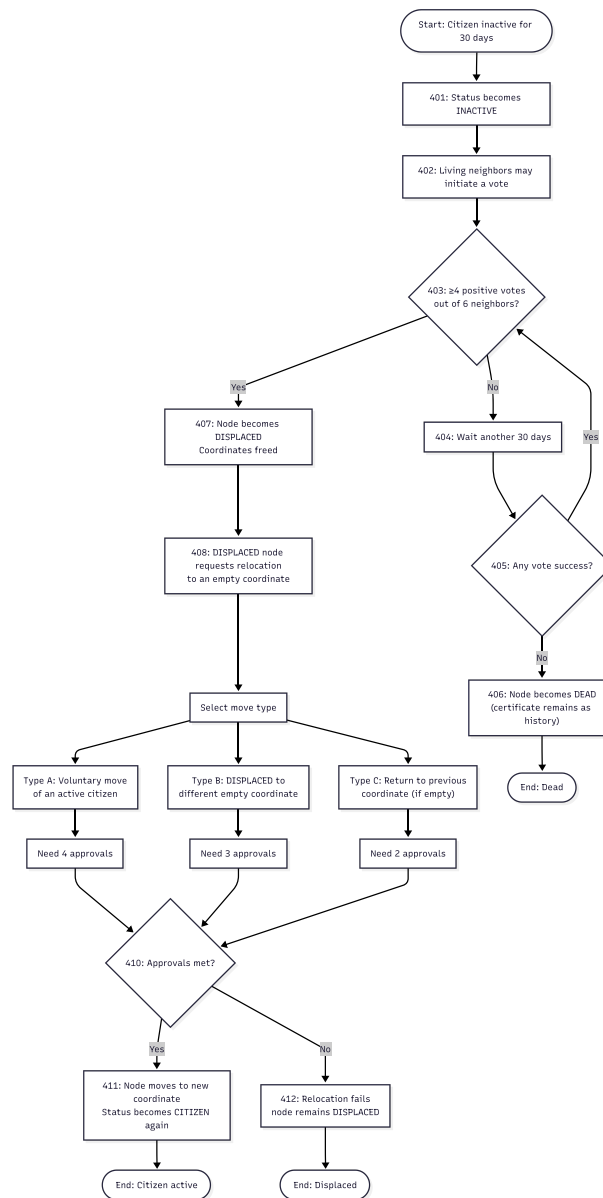
Fake news and rumors cannot collect three confirmations from trusted neighbors and never leave their local neighborhood.





3.3 Optical channels for simple propagation: For urgent information (e.g. earthquakes, security alerts) as well as to help the network spread, KANDO also provides optical channels. These optical channels are direct links between two non-affiliated users that bypass the 3 confirmation rule. The number of these channels is limited (e.g. 10 channels per user), they are only allowed for public and urgent content, and both parties must confirm the channel. Thus KANDO has a separate layer for simple propagation alongside the complex propagation layer (trust).

3.4 Three-stage peering (citizenship) protocol: A node (or Hive cell) goes through the following states: RESERVED: created by an invitation from an existing citizen with no neighbor conditions (the node occupies a coordinate but is not yet active). CANDIDATE: Promoted from reserved when all six direct neighbor slots are filled. (with any node type) the node becomes visible but cannot vote or create content. CITIZEN: Promoted from candidate when each of its six neighbors has at least six neighbors. This means that the second ring around the node is completely filled. A citizen has all the benefits of a full member. They can invite, issue endorsements, send content, and vote. Each node's citizen receives a non-transferable digital certificate (cNFT) A self-signed identity (DID) This certificate is permanently tied to the node's public key and can be used in all KANDO-compatible applications without revealing the real identity. Why is this protocol resistant to Sybil attacks? To create a fake citizen, an attacker must fill six neighborhood slots (and each of them must have six neighbors themselves). This requirement exponentially increases the cost of a large-scale attack.



3.5 Managing inactive nodes and relocations: A citizen who shows no activity for 30 days changes to the INACTIVE state. Live neighbors can start voting. The number of positive votes required depends on the number of live neighbors (four positive votes are required for six neighbors). If the vote is successful, the node changes to the DISPLACED state and its coordinates are released. If no successful vote is taken for an additional 30 days, the node goes to the DEAD state. Its certificate remains as a historical record but no longer grants active membership. A DISPLACED node can request relocation to an empty coordinate. The number of confirmations required from the neighbors of the destination coordinate depends

on the type of request: voluntary relocation of an active citizen and a DISPLACED node to a different empty coordinate and return to its previous coordinates if it is empty. If the confirmations are collected, the node is moved and its state changes back to CITIZEN, otherwise the move fails and the node remains in the DISPLACED state.

3-6 Two-layer architecture: overlay + physical mesh:

KANDO does not depend on a permanent internet connection and uses a (two-layer architecture) First layer (overlay network) = based on virtual coordinates (q, r) Neighbors are selected based on trust and common interests, not geographical location Two people on two different continents can be (virtual neighbors). Second layer (physical mesh network) = Bluetooth low energy Wi-Fi Direct or long-range technology (LoRa) is used for offline local communication in case of internet outage. So KANDO can use both online trust-based relationships and offline local communication. This network will be completely decentralized and has no central server. However, online relays are used for network stability.

3-7 Storage in DHT instead of server or blockchain

All node records, including coordinates, status, neighbor contributions, voting records, and certificates, are stored in a distributed hash table (DHT) like Kademlia. This solution makes the network without the need for a blockchain or gas fees, and users do not pay transaction fees.

4- Optional reward layer (Growth with gamification)

KANDO includes an optional and non-mandatory reward mechanism to encourage and grow the network. Users can earn points by inviting friends, endorsing content, voting, performing charitable behaviors, and participating in games. These points can increase the user's social reputation and are displayed as badges or numbers on their profile. No real money is involved in this process, and users can use the network without using the KANDO reward system. This layer is separate from the core protocol and does not affect the fundamental operation of the network.

5- Technical Effects and Benefits

Feature	Technical Effect
Hexagonal topology + Manhattan distance	$O(1)$ routing complexity
3-approval rule	60-80% bandwidth reduction, native spam filter
Co-eclosion protocol	Exponential Sybil resistance without PoW/PoS
Local voting and relocation	Self-healing, resilience to long-term internet outages
DHT storage	No blockchain, no gas fees – zero cost for ordinary users
Dual-layer architecture	Works offline via mesh networks, online via relays

6. Comparison with existing solutions

Feature	KANDO	BitChat / Briar	Nostr	Blockchain-based social networks
Offline mesh	 (Bluetooth, Wi-Fi Direct, LoRa)			
Gas fees	 zero	 zero	 zero	 (high)
Spam / misinformation filter	 (3-approval rule)			

Feature	KANDO	BitChat / Briar	Nostr	Blockchain-based social networks
Sybil resistance	✅ (co-eclosion)	❌	❌ (relies on optional PoW)	✅ (expensive)
Identity portability	✅ (cNFT / DID)	❌	✅ (public key)	✅ (NFT)
Scalable routing	✅ O(1)	❌ O(N)	❌ (relay-based)	🟡 (sharding needed)

7. Use cases:

Reporters and civil society activists: anonymous communication during internet outages, communication through content verified by multiple peers. Development and organization of social movements through complex contagion.

Relief organizations: coordination of rescue operations in disaster areas where communication infrastructure has been destroyed.

Decentralized application developers: building and developing on a trust layer that provides local identity and consensus.

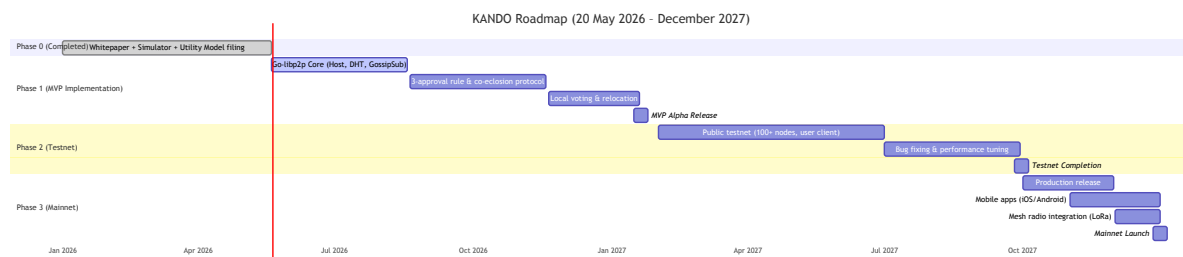
8. Intellectual property and open source

Patent status: An application model of the hexagonal topology and the rule of three verification and the co-binning protocol has been submitted to the Estonian Patent Office. However, all protocols and software developed or to be developed for KAND are or will be released under the GNU Affero General Public License v3 (AGPLv3). The source code of the protocol, simulator and documentation are freely available on GitHub.

Repository: <https://github.com/comfyuse/kando>

Website: <https://kandonet.com>

9. Roadmap



Phase Timeline Achievements

Phase 0 (Completed – before 20 May 2026)

- React simulator (proof of concept)
- Whitepaper (v1.0)
- Utility model filed with Estonian Patent Office

Phase 1 – MVP Implementation (20 May 2026 – 20 Jan 2027)

- **20 May – 20 Aug 2026:** Go-libp2p core (Host setup, Kademlia DHT, GossipSub)
- **20 Aug – 20 Nov 2026:** 3-approval rule, co-eclosion citizenship protocol, basic messaging
- **20 Nov 2026 – 20 Jan 2027:** Local voting, inactive node handling, relocation, testing
- **Milestone:** MVP Alpha release (20 Jan 2027)

Phase 2 – Public Testnet (Feb – Sep 2027)

- **Feb – Jul 2027:** Deploy testnet with 100+ volunteer nodes, user-friendly CLI client
- **Jul – Sep 2027:** Stress testing, bug fixing, performance tuning
- **Milestone:** Testnet stable (30 Sep 2027)

Phase 3 – Mainnet Launch (Oct – Dec 2027)

- **Oct 2027:** Production release of core protocol (AGPLv3)

- **Nov 2027:** Mobile apps (iOS/Android) using KANDO's DHT and mesh layers
- **Dec 2027:** Integration with LoRa radios for long-range offline mesh
- **Milestone:** Mainnet live (31 Dec 2027)

10. Call to Action

KANDO is a practical, scientific, and fully open source solution for a free and uncensored internet. We are looking for:

- Grant funding (e.g. NLnet, NGI) to accelerate the development of the Go libp2p implementation.
- Open source contributors (in the field of Go, cryptography, P2P networks).
- Early testers and usage partners (aid organizations, journalism networks).

Visit kandonet.com or join the discussion on GitHub.